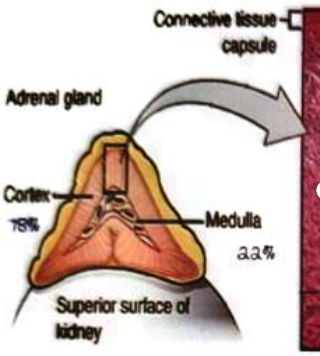
Www.Medicalstudyzone.com

ADRENAL GLAND

----- Active space --

Anatomy

00:00:56



Gross anatomy



Histopathology

CORTEX

- Zona glomerulosa :
  - Produces aldosterone (mineralocorticoid).
  - Contains stem cells For regeneration of cortical cells.
- Zona fasciculata (50 %) : Produces cortisol (Glucocorticoid).
- Zona reticularis : Produces adrenal sex steroids

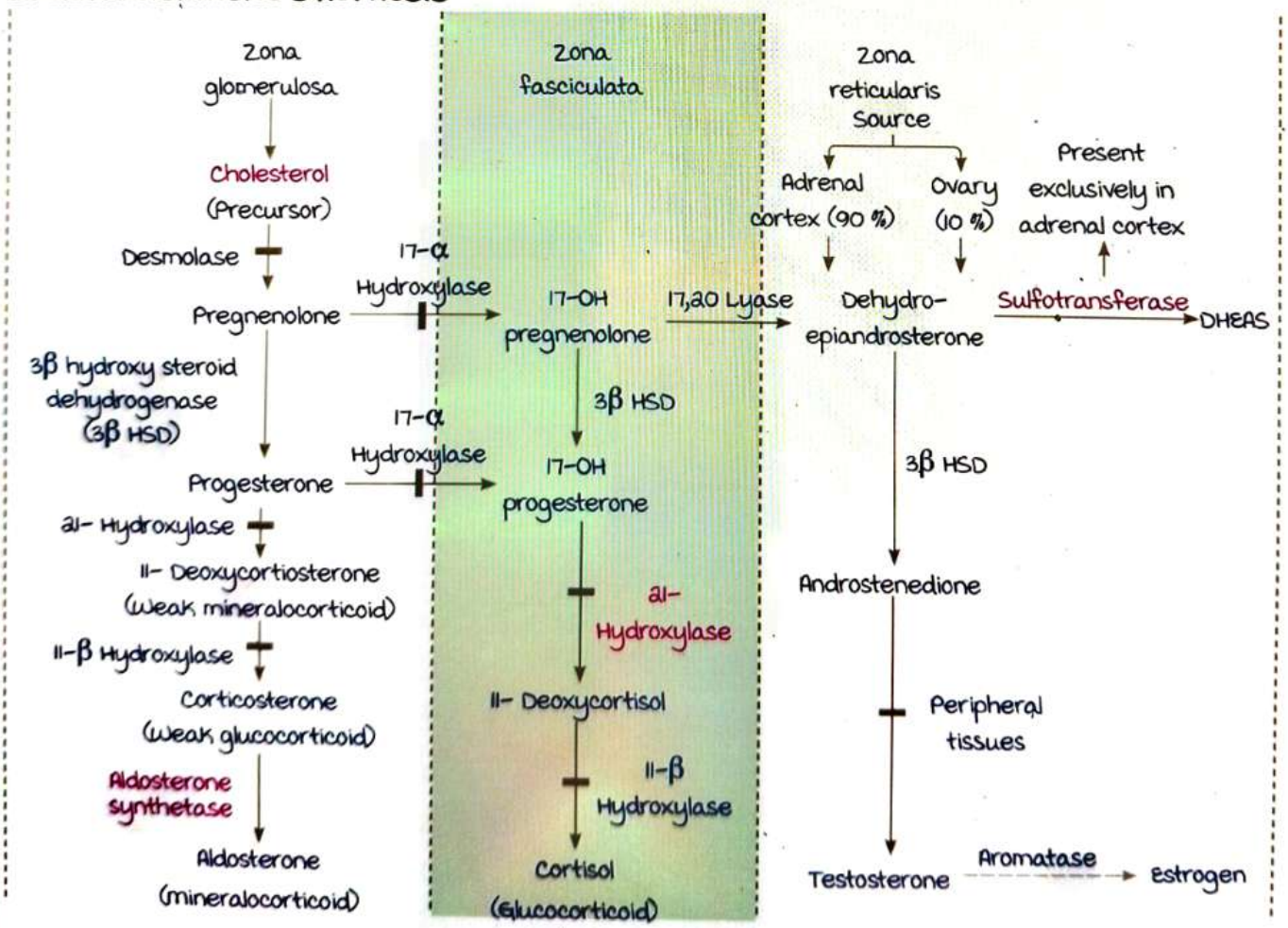
MEDULLA

- Medulla : Produces catecholamines.
  - Epinephrine.
  - Norepinephrine.
  - Dopamine.

Physiology

00:05:05

STEROID HORMONE SYNTHESIS



Www.Medicalstudyzone.com

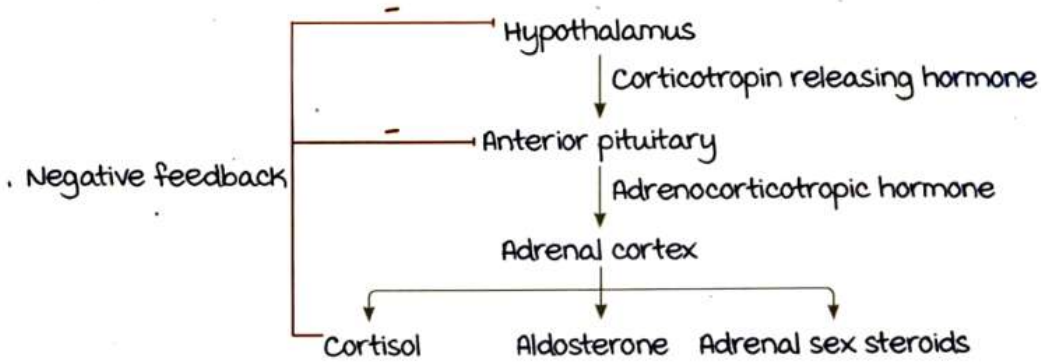
----- Active space -----

Deficiency of enzymes involved → Congenital adrenal hyperplasia.

Note : Major source of testosterone → Testis.

Regulation of hormone synthesis :

Hypothalamic-pituitary-adrenal (HPA axis) :



Note : Adrenal medulla is not under ACTH control.

Adrenocortical Hormones

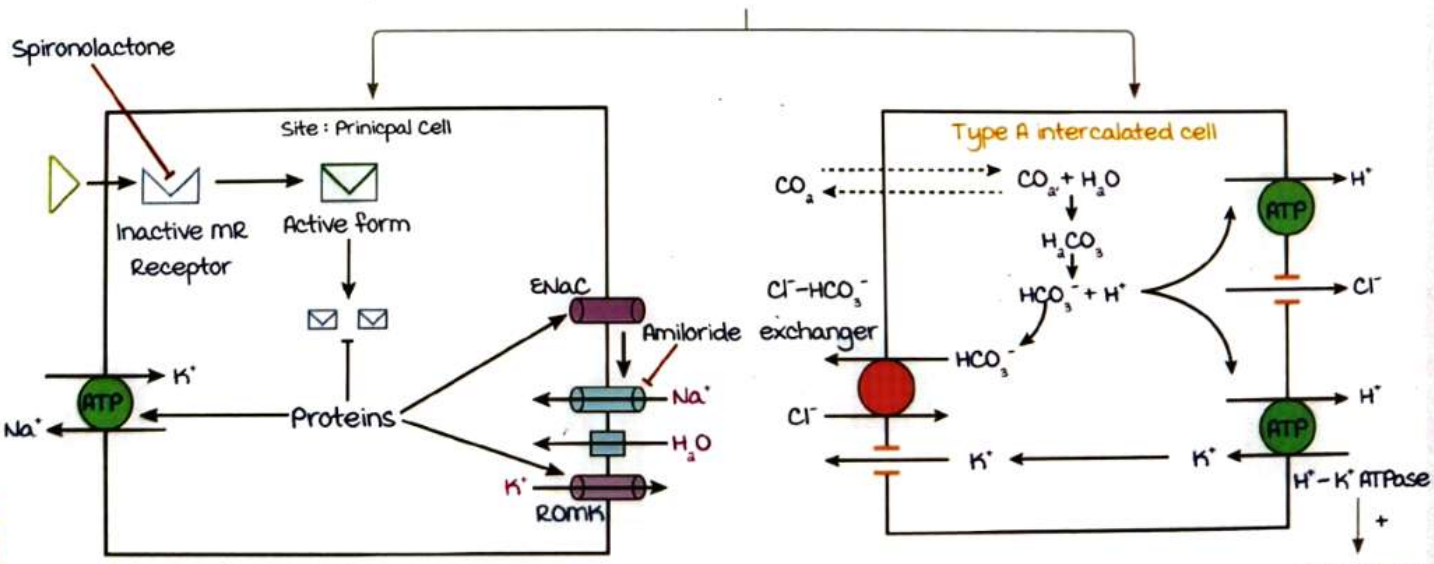
00:16:05

ALDOSTERONE :

- Steroid hormone (Life saving).
- 60 % : Present in bound form with binding proteins.
- $t_{1/2}$: 20 mins.

Mechanism of action :

Kidneys : In collecting duct of nephron :



ENaC : Epithelial Na^+ channel.

ROMK : Renal Outer Medullary Channel

MR : Mineralocorticoid Receptor

MR sites : Kidneys, colon, hippocampus, salivary glands, sweat glands.

Feedback

Www.Medicalstudyzone.com

Factors regulating aldosterone synthesis :

----- Active space -----

- Hyperkalemia → Stimulates aldosterone release.
- ACTH → ↑ aldosterone transiently.

Disorders :

Hyperaldosteronism :

1. Conn's syndrome :
   - Etiology : Aldosterone producing adrenal adenomas.
   - Clinical features :
     a. Metabolic alkalosis.
     b. Hypokalemia → Muscle cramps, Interstitial atony, Nephropathy
     c. Hypertension : Na^+ & H_2O retention → ↑ ECF → ↑ BP but no edema.

Aldosterone escape :

Pressure diuresis (D/t ↑ BP) + Atrial natriuretic peptide : Natriuresis → Na^+ & H_2O excretion (in spite of ↑ aldosterone) → No edema

2. Glucocorticoid remediable hyperaldosteronism :
   - Pathophysiology :
     - Aldosterone synthase + 11 β-hydroxylase → Hybrid enzyme
     - ↑ ACTH → Sustained release of aldosterone (D/t hybrid enzyme).
   - Treatment : Dexamethasone (Inhibits ACTH).

Hypoaldosteronism :

1. Addison's disease (Adrenal insufficiency).
   - Features :
     a. Hyponatremia.
     b. Postural hypotension.
     c. Hyperkalemia.
     d. Microcardia.

CORTISOL :

- 90–95 % bound to cortisol binding globulin (CBG)/transcortin.
- $t_{1/2}$: 60–90 mins.

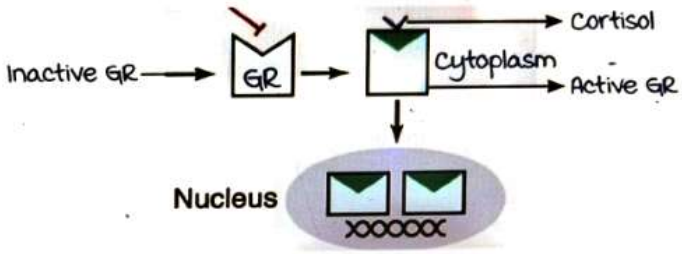
Feedback

Www.Medicalstudyzone.com

----- Active space -----

Mechanism of action :

Glucocorticoid receptor (GR) :



Functions :

	Physiological (Normal cortisol)	Pathological : Cushings syndrome (cortisol excess)
1. Protein metabolism	Muscle : Proteolysis → Alanine (Catabolic) Liver : Synthesis of plasma proteins (Anabolic)	Muscle weakness Proximal myopathy (Steroid myopathy)
2. Carbohydrate metabolism	Gluconeogenesis → Hyperglycemia (Alanine → Glucose)	Glucose intolerance Diabetes mellitus (Adrenal diabetes)
3. Lipid metabolism	Lipolysis → ↑ free fatty acids	Dyslipidemia 
4. Immune system	Universal anti-inflammatory agent : Inhibits → Leukotrienes Inhibits → Phospholipases A_2	↑ susceptibility to infections
5. Blood cells	↓ : Eosinophils, basophils & lymphocytes. ↑ : RBCs, neutrophils & platelets.	Eosinopenia Hypercoagulable state
6. Nervous system	↑ appetite → Weight gain. Alters mood & behavior.	Irritability Psychosis
7. Heart	↑ cardiac output ↑ vascular tone	Hypertension Atherosclerosis
8. Kidney	↑ GFR ↑ Ca^{2+} excretion	Hypercalciuria → Renal stones
9. Bone & connective tissues	Inhibits → Fibroblast proliferation Inhibits → Collagen formation Promotes : Bone resorption.	
10. GIT	↑ gastric acid production	Peptic ulcer
11. Reproduction	Inhibits GnRH (Negative regulator of reproduction)	Amenorrhea Loss of libido Infertility

Feedback

Www.Medicalstudyzone.com

12. Effect on catecholamines :

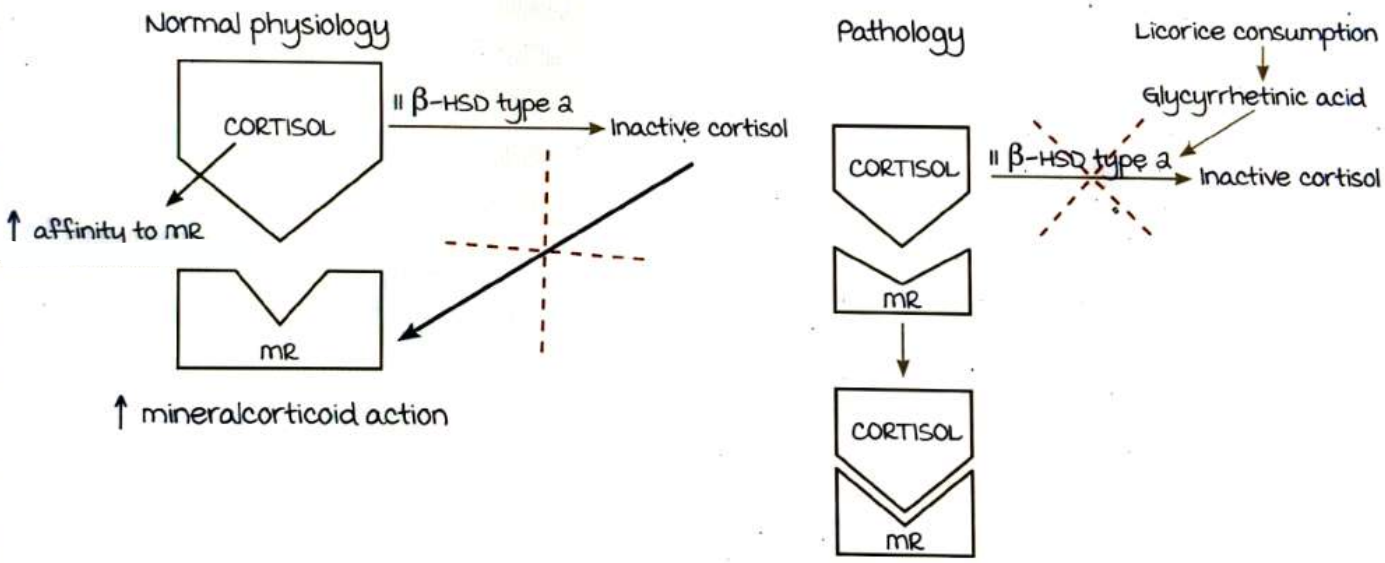
----- Active space -----

Catecholamines —(In presence of cortisol)→ Enhanced actions :
- Lipolysis.
- Bronchodilation.

Other diseases :

Addisons disease :
- Cortisol deficiency.
  1. Hypoglycemia.
  2. Fatigue.
  3. Eosinophilia.
  4. Lymphocytosis.

Syndrome of apparent mineralocorticoid excess (SAME) :



SEX STEROIDS

Hormones :
1. Dehydroepiandrosterone (DHEA).
2. DHEA sulfate (DHEAS) : most circulating hormone in humans.
3. Androstenedione.
4. Testosterone (Small amounts).

Functions :
1. Axillary hair growth.
2. Pubic hair growth.
3. Libido.

Feedback

Www.Medicalstudyzone.com

----- Active space -----

Disorders :

Causes :

1. Adrenocortical tumors
2. Congenital adrenal hyperplasia } ↑ adrenal androgens
3. Adrenogenital syndrome :

Clinical features

- Females (more severe) → Pseudohermaphroditism :
  - Beard growth.
  - Deep voice.
  - Baldness.
  - Hirsutism.
  - Clitoromegaly.
- Males (Prepubertal time) → Rapid development of male sex organs

Feedback